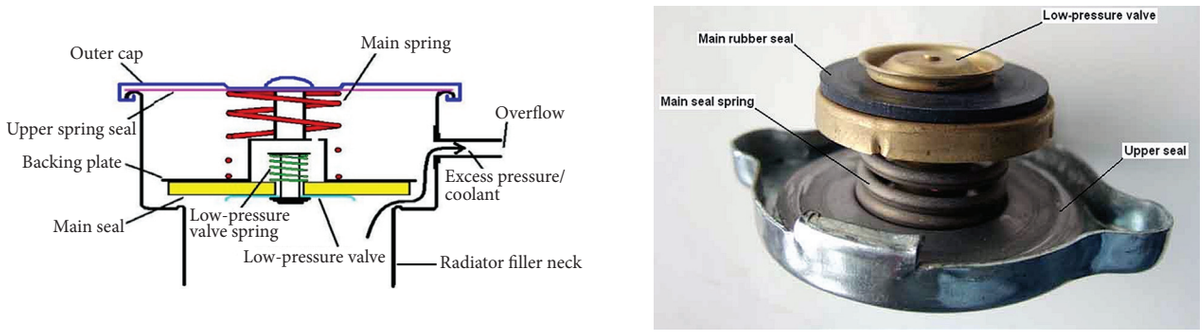


Figure 7.4 Pressure Cap

7.4 PRESSURE CAP

The component used to close the upper tank of the radiator is called as the radiator pressure cap. This cap prevents the flow of evaporated water (steam) outside from the upper tank. It also helps in avoiding flow of water outside the tank when the vehicle is traveling on any bumps and pot holes. In cold climatic countries water inside the radiator gets cooled naturally when the engine is at rest. In such condition vacuum may occur inside the tank. In such situations the exterior air comes inside the tank and replaces the vacuum area. The pressure in the pressure cap is released from the pressure valve by opening the pressure cap when the pressure inside the water tank is increased. When water is cooled the vacuum vale is opened and air is allowed to enter in to the radiator for protecting it.

7.5 THERMOSTAT

A specified temperature must be maintained in the cooling system to operate the engine safely and efficiently. This temperature is generally in the order of 70 deg to 80 deg for petrol engine and 80 deg to 85 deg for diesel engines. To maintain the temperature a thermostat arrangement is used. It blocks the coolant supply until the engine is warmed up. The thermostat valve is placed in between the engine and upper tank of the radiator. Thermostat housing consists of an inlet and outlet valve. Inside the housing the thermostat is placed. There is a bypass line arrangement for passing cooling water after the engine is started and until it reaches the required temperature. Cooling water starts to flow through the tubes by opening the thermostat valve when the engine reaches its specified temperature and the cooling water is sent to the upper tank in the radiator.

Basically there are two types of thermostat valves used in automobiles, they are

1. Bellows type thermostat
2. Wax type thermostat

7.5.1 Bellow Type Thermostat

In this type of thermostat, there is a frame in the upper side and valve in the lower side which is attached to the bellows. Bellows are filled with either alcohol or acetone which is easily evaporating chemicals. These chemicals have lower boiling points such as 70 deg to 80 deg. When the engine's coolant(water)

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